

Rapid Response Information System (RRIS)

An AI-Assisted Emergency Incident Information System

Project Overview and Motivation:

During emergencies such as fires, vehicle crashes, severe weather, medical emergencies, and hazardous-material incidents, information can change rapidly. Emergency personnel may receive information from 911 dispatch, field reports, weather services, traffic systems, and other sources. These reports can be incomplete, duplicated, delayed, or presented in different formats.

The problem is not necessarily a lack of information. The challenge is determining what information belongs to the same incident, what is most important, how reliable it is, and what has changed.

Our project proposes the Rapid Response Information System (RRIS), an AI-assisted system that consolidates emergency information into one clear incident picture for first responders. The system will begin with information originating at the 911/dispatch level and combine it with supporting information such as weather, traffic, roadway conditions, and infrastructure impacts.

The project will use Lubbock, Texas, as its demonstration area. This provides a realistic environment for testing emergency scenarios while keeping the project geographically manageable. Actual private 911 or CAD information will not be accessed; instead, the team will create realistic simulated emergency reports and combine them with appropriate public data.

RRIS will follow a Human-Centered AI approach: AI provides recommendations, a human confirms or adjusts them, and responders make operational decisions. The system will not replace dispatchers or determine how individual agencies should respond.

Problem Statement and Intended Users:

During a major emergency, several reports may describe the same event. For example, a vehicle crash could generate a 911 report, an EMS update, a fire department report, a traffic alert, and a weather warning. Without a system that connects these reports, users may see multiple alerts for what is actually one incident.

RRIS will address this problem by combining related reports into a unified incident record. The system will also consider supporting factors that may affect the incident, including injuries, active hazards, weather, traffic, accessibility, infrastructure impacts, and the age of the information.

The intended users are first responders and emergency personnel, including police, firefighters, EMS, emergency transportation personnel, dispatchers, and emergency-management personnel.

Rather than creating separate police, fire, and EMS interfaces, RRIS will provide a shared incident view. This keeps the project feasible and avoids assuming that different agencies use identical procedures or standard operating procedures. Each agency can use the information according to its own training and policies.

The system is intended to answer questions such as:

- * What happened?
- * Where is it happening?
- * How serious is it?
- * What hazards are present?
- * What supporting conditions could affect the response?
- * What has changed since the incident was first reported?
- * How confident is the system in the available information?

Proposed System and Functionality:

RRIS will process information through the following general workflow:

911/Dispatch → Reports → Incident Fusion → AI Extraction and Classification → Priority Scoring → Human Review → Unified Incident View

Incoming reports will first be normalized into a common structure. The system will then determine whether multiple reports may describe the same incident. For example, reports describing a vehicle collision at the same location within a short period could be combined into one incident instead of appearing as separate alerts.

The system will classify incidents using eight primary categories:

- | | |
|-----------------------------|--------------------------------|
| 1. Fire / Explosion | 5. Hazardous Materials |
| 2. Medical / Mass Casualty | 6. Infrastructure / Utility |
| 3. Transportation | 7. Public Safety / Security |
| 4. Weather / Natural Hazard | 8. Rescue / Special Operations |

Only a practical subset of subcategories will be implemented for the demonstration.

RRIS will use a hybrid AI approach. An AI/LLM component may extract information from narrative reports, identify relevant details, and summarize multiple reports. Transparent rule-based logic will then be used for classification and priority scoring. This separation allows the team to explain how recommendations are produced.

Priority will be based on several factors:

Factor Weight

Life safety / injuries	30%	Accessibility / transportation impact	10%
Active hazards	20%	Infrastructure / community impact	10%
Incident severity / scope	15%	Confidence adjustment	5%
Time sensitivity	10%		

An important design decision is to keep severity and confidence separate. A potentially catastrophic incident may have high severity but low confidence if it has not yet been confirmed. RRIS will communicate both rather than treating an uncertain report as confirmed.

The system will also provide an explanation for its priority recommendation. For example, it might identify multiple injuries, an active fuel leak, severe weather, and a major roadway obstruction as factors contributing to a high-priority recommendation.

A human dispatcher will then be able to confirm, adjust, or reject the recommendation. This keeps the human decision-maker at the center of the system.

The final interface will provide a unified incident view showing the incident location, classification, priority, confidence, injuries, hazards, weather, traffic, supporting information, source reports, and recent changes. A map will provide geographic context.

Data, Inputs, and Tools:

The project will use simulated emergency information and publicly available supporting data.

The team will create simulated 911/dispatch reports using Python. These reports will contain realistic information such as incident descriptions, locations, timestamps, injuries, hazards, and updates. Multiple reports will sometimes describe the same incident so that incident fusion can be tested.

Public information will be used to provide additional context. Weather information from the National Weather Service is one potential source for the Lubbock demonstration area. Other appropriate public transportation, geographic, or infrastructure information may also be incorporated.

The proposed technology stack includes Python, pandas, NumPy, scikit-learn, FastAPI, React, Leaflet.js, SQLite/PostgreSQL, GitHub, and Jupyter/Anaconda. A lightweight LLM API may be used for information extraction and summarization.

The team will prioritize a functioning prototype over unnecessary technical complexity.

Timeline and Milestones:

Phase 1 — Proposal and Planning

Due August 30

- * Finalize proposal and project scope
- * Establish GitHub repository
- * Define data structure

- * Select initial incident categories

- * Define scoring framework

- * Identify public data sources

- * Develop initial scenarios

Phase 2 — Data and Intelligence

- * Create simulated emergency data

- * Implement data normalization

- * Build classification rules

- * Develop priority scoring

- * Begin AI-assisted extraction

Phase 3 — Incident Fusion and Updates

- * Combine related reports

- * Add confidence and source information

- * Track incident changes

- * Test changing priority levels

Phase 4 — Interface and Integration

- * Develop unified dashboard

- * Add map visualization

- * Integrate scoring and classification

- * Add explainability

- * Add human confirmation/adjustment

Final Phase — Testing and Presentation

- * Test realistic Lubbock scenarios

- * Evaluate system performance

- * Document limitations

- * Complete final demonstration and presentation

Team Roles and Responsibilities:

Derrick Hawkins — Project Coordinator / Research / Simulated Data

Derrick will coordinate the project, conduct research, develop simulated 911/dispatch scenarios, research potential public data sources, assist with integration, and contribute to documentation and testing.

Benjamin Duske — Architecture / UI/UX

Benjamin will focus on system architecture, frontend development, dashboard design, mapping, backend integration, and user experience.

Marc Griffin — Data / Classification / Priority Scoring

Marc will focus on data structure, normalization, incident classification, incident fusion, priority scoring, confidence handling, and explainability.

Nicholas Kelly — Research/Priority Scoring/ UI/UX/

Nicholas will support research, refine priority scoring models, and collaborate with UI/UX to design how severity, confidence, and explainability are visually represented in the dashboard.

All team members will participate in testing, documentation, major design decisions, and the final presentation.

Risks, Limitations, and Ethical Considerations

AI-generated information can be incorrect. RRIS will therefore treat AI output as a recommendation rather than an authoritative decision. Human confirmation will remain part of the workflow.

Incident fusion may also incorrectly combine separate events or fail to recognize related reports. The system will retain source information and timestamps so users can review the underlying information.

Another concern is information overload. A system that simply adds more information could make emergency responses more difficult. RRIS will therefore focus on prioritization, concise explanations, supporting context, and showing what has changed.

Privacy is also important because real emergency reports may contain sensitive personal information. The project will use simulated, anonymized, or publicly available data rather than private emergency records.

Finally, RRIS will not prescribe actions for individual agencies. Different departments may have different SOPs and response procedures. The system will provide a shared incident picture while allowing trained personnel to make their own operational decisions.

The prototype will be an academic proof-of-concept and will not have the reliability, cybersecurity, redundancy, or certification required for a production emergency-response system.

Conclusion

RRIS will demonstrate how Human-Centered AI can help organize rapidly changing emergency information without removing humans from the decision-making process.

The system will begin with information originating at the 911/dispatch level, combine multiple reports and supporting information, identify related reports, classify incidents, calculate transparent priority recommendations, and present the results through a unified incident view.

By focusing on one shared picture for all first responders rather than separate agency-specific systems, the project remains feasible while addressing a meaningful real-world problem.

The central goal of RRIS is to:

Give emergency personnel one clear, current, and explainable picture of an incident while keeping the human decision-maker in control.

References

National Weather Service. Lubbock, Texas Weather Information and Local Resources.
<https://www.weather.gov/lub/>